

Journal of Graduate Medical Education

Training Through Turbulence: Why Medical Education Must Build Intentionally Supportive, Historically Grounded Learning Cultures

--Manuscript Draft--

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Short Title:	Training Through Turbulence
Article Type:	Perspectives
Keywords:	medical education; learning culture; supportive learning environments; psychological safety; clinical teaching

Author Response

Introduction

Thank you for the very helpful feedback. We have narrowed the central theme of the article to focus on the historical grounding through out the paper and added a table with concrete and practical steps. As a result, we have made some changes that are highlighted in the version with tracked changes, and we will attempt to highlight them in response to reviewer comments as well.

Previous word count (abstract and manuscript): 812

Revised word count (abstract and manuscript): 933

Reviewer 1 Comments

1. I concur with the recommendation for a major revision before the manuscript proceeds to external review. While the topic is promising, the current draft lacks sufficient specificity, and several of the ideas are still too preliminary to support the conclusions drawn. I would encourage the authors to narrow their focus to one central theme, present the supporting evidence more clearly, and then offer concrete, evidence-based recommendations aligned with that focus.

Additionally, incorporating a figure, such as a conceptual model, workflow, or summary visualization, would strengthen the manuscript and enhance its clarity and impact.

Thank you for this suggestion, which we very much took to heart. You will see that we primarily focused our revision on how a historical perspective can be a guiding principle to navigate bleak times and provide a new Table 1 with more specific actionable guidance for how to apply this perspective at the level of teacher-student and organizationally.

Location in text (page and line numbers): See track changes and new Table 1.

Page 1 – paragraph 3 has been added:

“Patient-centered care is a prerequisite that must be prioritized based on patient needs, clinical indication, legal permissibility, and what can be realistically achieved. Faculty must model and teach patient-centered reasoning, ensuring that trainees are not managing uncertainty alone and know when to escalate issues to legal, ethics, or risk management teams. The impact of organizational decisions on patient care should be transparently communicated, and moral distress should be acknowledged as treatment and care responsibility is shared across teams. Intentionally creating these reflective and supportive spaces that center around the experience of care providers (e.g., Balint groups) is key as clinicians strive to deliver the best care for their patients.”

Page 3- paragraph 2 has also been reworked to align with the historical perspective:

“Discuss the historical evolution of organizational cultures in academic medicine.

Throughout history, the most transformative growth has resulted when structural changes at organizational level have fostered the clinician’s ability to deliver humanistic care under extreme duress. Lessons learned from wartime and pandemic crises that led to forced systemic solutions highlight the importance of supportive and communicative organizational cultures rather than individual fortitude¹⁶. While individual resilience and mentorship are indispensable, they cannot flourish in a vacuum. The emotional labor expected of clinicians and trainees today is inseparable from the environment in which they work. Cultivating these humanistic attributes in

organizations is equally important as in individuals to counter the negative effects of healthcare evolution.¹⁷

Reviewer 2 Comments

1. The 'what/how to do' is somewhat vague. Perhaps adding some specifics in text, or a table to examples (resident/fellow stress - potential faculty response - potential insitutional response) for some common issues, different specialties (JGME read by all specialteis)

While much will apply to all levels of training, the most useful examples pertain to GJGME

As with Reviewer 1's comments above, which were highly congruent with yours, we greatly appreciated this advice to make the commentary more targeted and actionable. You will see track changes throughout the document, as well as a new Table 1, showing this greater specificity and actionability – all centered around the guiding principle of “lessons from history.”

Location in text (page and line numbers): Changes are quite extensive. Please see track changes and new Table 1.

2. Actions/ recommendations should be supported by evidence (studies, theories, concensus - even outside medicine, if none available in medicine) or by authors' expertise/experience.

Thank you so much for this very helpful suggestion. Please refer to the expanded bibliography and newly annotated Table 1 for a broad expansion of evidence-based suggestions for supporting learners.

Location in text: Table 1, bibliography.

Training Through Turbulence: Why Medical Education Must Build Intentionally Supportive, Historically Grounded Learning Cultures

The current moment in medical education is marked by compounding pressures: lingering effects of the COVID-19 pandemic, workforce shortages, financial strain, and ongoing moral distress associated with providing care in stretched-thin systems^{1,2}. Political polarization, societal inequities, and rising mistrust in science further heighten stress for both clinicians and learners. These realities intensify the emotional and moral challenges of learning to care – and can make the humanistic ideals that drew many to medicine feel out of reach³⁻⁷.

Our humanistic ideals remain critical, central, and immutable, as is our duty to uphold them even in the face of the current political and geopolitical climate. A foundation of universal values, which requires institutional/organizational support, can foster clinicians' ability to lead with ethical, professional, and moral integrity even during challenging circumstances⁸. Although it appears intuitive, trainees need explicit reminders to take care of the patient in front of you, do what you can, don't dwell on what you can't.

Patient-centered care is a prerequisite that must be prioritized based on patient needs, clinical indication, legal permissibility, and what can be realistically achieved. Faculty must model and teach patient-centered reasoning, ensuring that trainees are not managing uncertainty alone and know when to escalate issues to legal, ethics, or risk management teams. The impact of organizational decisions on patient care should be transparently communicated, and moral distress should be acknowledged as treatment and care responsibility is shared across teams.

Intentionally creating these reflective and supportive spaces that center around the experience of care providers (e.g., Balint groups) is key as clinicians strive to deliver the best care for their patients.

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6 clinicians and trainees but in meeting them with clarity, solidarity, and purpose. Below, we offer
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8 suggestions for navigating these tumultuous times, using our humanistic values as the central
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10 guiding principle.
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17 **Provide historical perspective.** As reflected in their responses to epidemics, new illnesses,
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19 wartime medicine, and scarcity of resources, most physicians, be they learners or teachers,
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21 ultimately navigate a path out of darkness. Current dilemmas are shaped by context and systemic
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25 this longitudinal perspective and would likely benefit from historical recollections and narratives
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29 modeling resilience, and authentic disclosure on the part of mentors, trainees remain vulnerable
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31 to the fallout of their own high expectations. Some trainees even seek out geographical
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33 relocation to escape the angst of value misalignment.
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43 **Acknowledge the uncertainty of our times, validate the stress it generates, and balance the**
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45 **emotional toll it brings with hope.** The history of medicine is rich with examples of resilience.
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15 **Senior clinicians should model and articulate how they are preserving their values and**
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16 a group with a focus on promoting moral agency, teamwork, and resilience-building activities.
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18 Despite external pressures and the outward-facing measures they must take, it is incumbent on
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22 centered care.
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26 To translate these conceptual values into practice, **Table 1** offers practical guidance and
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28 strategies that can be translated and tailored across specialties.
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36 ***In Conclusion***
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39 Medicine has always evolved through adversity. Even in these uncertain times, with the guiding
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41 principle of patient-centered care and a community united around this principle, we can both
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43 withstand and transcend the pressures of the present.
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47 **References:**
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51 1. Singer T, Klimecki OM. Empathy and compassion. *Curr Biol.* 2014;24(18):R875-R878.
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53 2. Graves J, Joyce C, Ahmed M, Hegazi I. Compassion fatigue in medical students and
54 recent medical graduates: a scoping review. *Teach Learn Med.* 2025;:1-20.
55
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3. Mantri S, Song YK, Lawson JM, Berger EJ, Koenig HG. Moral injury and burnout in health care professionals during the COVID-19 pandemic. *J Nerv Ment Dis.* 2021;209(10):720-726.
4. Beadle ES, Walecka A, Sangam AV, Moorhouse J, Winter M, Munro Wild H, et al. Triggers and factors associated with moral distress and moral injury in health and social care workers: a systematic review of qualitative studies. *PLoS One.* 2024;19(6):e0303013.
5. Chandrabhatla T, Asgedom H, Gaudiano ZP, de Avila L, Roach KL, Venkatesan C, et al. Second victim experiences and moral injury as predictors of hospitalist burnout before and during the COVID-19 pandemic. *PLoS One.* 2022;17(10):e0275494.
6. Mangurian C, Fitelson E, Devlin M, Pumar M, Epel E, Dahiya P, et al. Envisioning the future of well-being efforts for health care workers—successes and lessons learned from the COVID-19 pandemic. *JAMA Psychiatry.* 2023;80(9):962-967.
7. Hughes MT, Rushton CH. Ethics and well-being: the health professions and the COVID-19 pandemic. *Acad Med.* 2022;97(3S):S98-S103.
8. Rider EA, Kurtz S, Slade D, Longmaid HE III, Ho MJ, Pun JK, et al. The International Charter for Human Values in Healthcare: an interprofessional global collaboration to enhance values and communication in healthcare. *Patient Educ Couns.* 2014;96(3):273-280.
9. McLean M, Jha V, Sandars J. Professionalism under fire: conflict, war and epidemics. *Med Teach.* 2015;37(9):831-836.
10. Lerner BH, Caplan AL. Judging the past: how history should inform bioethics. *Ann Intern Med.* 2016;165(9):675.

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2
3
4 11. Coria AL, Rabin TL, Rule AR, Haq H, Hudspeth JC, Ratner L, et al. Global health crisis,
5
6 global health response: how global health experiences prepared North American
7
8 physicians for the COVID-19 pandemic. *J Gen Intern Med*. 2022;37(1):217-221.
9
10
11 12. Edmondson AC. *The Fearless Organization: Creating Psychological Safety in the*
12
13 *Workplace for Learning, Innovation, and Growth*. Hoboken, NJ: John Wiley & Sons;
14
15 2018.
16
17
18 13. Whitaker RC, Payne GB, O'Neill MA, Brennan MM, Herman AN, Dearth-Wesley T, et
19
20 al. Trauma-informed undergraduate medical education: a pathway to flourishing with
21
22 adversity by enhancing psychological safety. *Perspect Med Educ*. 2024;13(1):324.
23
24
25 14. Toubassi D, Schenker C, Roberts M, Forte M. Professional identity formation: linking
26
27 meaning to well-being. *Adv Health Sci Educ Theory Pract*. 2023;28(1):305-318.
28
29
30 15. Jones DS, Greene JA, Duffin J, Warner JH. Making the case for history in medical
31
32 education. *J Hist Med Allied Sci*. 2015;70(4):623-652. doi:10.1093/jhmas/jru026
33
34
35 16. Curtin M, Richards HL, Fortune DG. Resilience among health care workers while
36
37 working during a pandemic: a systematic review and meta-synthesis of qualitative
38
39 studies. *Clin Psychol Rev*. 2022.
40
41
42 17. Rider EA, Gilligan MC, Osterberg LG, Litzelman DK, Plews-Ogan M, Weil AB, et al.
43
44 Healthcare at the crossroads: the need to shape an organizational culture of humanistic
45
46 teaching and practice. *J Gen Intern Med*. 2018;33(7):1092-1099.
47
48
49 18. Bryan CS, Longo LD. Perspective: teaching and mentoring the history of medicine: an
50
51 Oslerian perspective. *Acad Med*. 2013.
52
53
54 19. Teo R, Grosser R, Thuppall H, Statter MB. "Sip & Share": building resilience in surgery
55
56 residency through moral distress rounds. *J Surg Educ*. 2023.
57
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2
3
4 20. Metselaar S, Molewijk B. Fostering moral resilience through moral case deliberation.
5
6
7 *Nurs Ethics*. 2023.
8
- 9 21. Feingold JH, Kaplan CA, Hart A, et al. We Get by With a Little Help From Our PEERS:
10
11 The Practice Enhancement, Engagement, Resilience, and Support Program for Building
12
13 Community and Well-Being in Medical Education. *Acad Med*. 2022;97(6):858-862.
14
15
16 doi:10.1097/ACM.0000000000004669
17
- 18 22. Horvath AJ, Shegufta FA, Chairilsyah MR, Wikanta K, Waidyatillake NT, Reid K. The
19
20 effectiveness of interventions designed to enhance and maintain medical student
21
22
23 wellbeing: a systematic review. *BMC Med Educ*. 2026;26(1):667. Published 2026 Mar 7.
24
25
26 doi:10.1186/s12909-025-08460-5
27
- 28 23. Thomas LR, Brigham T, Shanafelt T. Residency and Fellowship: Fostering Physician
29
30 Well-being Over the Career Life Cycle. *Mayo Clin Proc*. 2025;100(9):1649-1659.
31
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34 doi:10.1016/j.mayocp.2025.05.024
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	How faculty can respond	How institutions can respond
Provide historical perspective	• Share historical narratives of physician resilience during past crises (1918 influenza, AIDS epidemic, Chernobyl disaster, SARS) through structured discussions or curriculum segments¹⁸ • Supervision on aspects of “current landscape” • Update on how and when policy changes affect care • Structured debrief sessions such a sip and share rounds (structured 8-step methodology)¹⁹, or moral case deliberation²⁰	• Build safe forums where trainees and faculty can ask questions without fear of being judged, for their views and political leaning • Formal integration of history of medicine segments in major course introductions and sponsor history clubs • Websites and apps with policy availability • Open forums where leadership can share how they have navigated such challenges in the past
Acknowledge uncertainty, validate stress, and balance emotional toll with hope	• Acknowledge uncertainty and what the team will do meanwhile • Use brief structured check-ins after difficult cases: “What happened? What felt ethically hard? What	• Institution based programs for individual and collective support and debriefing • Provide timely access to confidential mental health services

	support is needed? What follow-up would be needed later?” • Longitudinal peer support groups (PEERS Model: Practice, Enhancement, Engagement, Resilience and Support) ²¹ • CBT and mindfulness based interventions, gratitude practices, including online options²²	outside the trainee and faculty evaluation chain with protected time to attend • Protect time for debriefing after challenging cases and/or legally or ethically fraught encounters • Schedule community building activities
Mentors and educators' role	• Schedule regular, expected mentorship proactively • Address professional identity development during the current climate • Teach “permission to escalate” explicitly: trainees should know they are not expected to absorb moral conflict alone • Separate coaching and mentorship from evaluation when possible • Instruct trainees on: who to call, how to escalate, what	• Build layered mentorship: faculty mentor, peer mentor, and near-peer mentor • Train faculty in recognizing distress, moral injury, and specialty-specific burnout patterns • Protect faculty time for mentorship • Provide faculty development training in creating psychological safety, and vulnerability modeling²³

	documentation standards apply, and where to access ethics/legal support	• Ensure faculty are provided up to date guidance about policy changes, institutional guidance and support available
Attend to organizational culture	• Explicitly acknowledge what system level issues and failings are • Reduce avoidable administrative friction • Share workload transparently across teams • Model professional behavior and values that collectively impact institutional culture	• Foster resilience with structural support • Monitor rotation-level workload, call burden, EMR burden, and staffing gaps • Align policies and procedures with patient centered care considering constraints for consistency and transparency

Training Through Turbulence: Why Medical Education Must Build Intentionally

Supportive, Historically-Grounded Learning Cultures

The current moment in medical education is marked by compounding pressures: lingering effects of the COVID-19 pandemic, workforce shortages, financial strain, and ongoing moral distress associated with providing care in stretched-thin systems^{1,2}. Political polarization, societal inequities, and rising mistrust in science further heighten stress for both clinicians and learners. These realities intensify the emotional and moral challenges of learning to care – and can make the humanistic ideals that drew many to medicine feel out of reach³⁻⁷.

~~Focus on the best interests of the patient.~~ Our humanistic ideals remain critical, central, and immutable, as is our duty to uphold them even in the face of the current political and geopolitical climate. A foundation of universal values, which requires institutional/organizational support, can foster clinicians' ability to lead with ethical, professional, and moral integrity even during challenging circumstances¹¹⁸. Although it appears intuitive, trainees need explicit reminders to take care of the patient in front of you, do what you can, don't dwell on what you can't.

Patient-centered care is a prerequisite that must be prioritized based on patient needs, clinical indication, legal permissibility, and what can be realistically achieved. Faculty must model and teach patient-centered reasoning, ensuring that trainees are not managing uncertainty alone and know when to escalate issues to legal, ethics, or risk management teams. The impact of organizational decisions on patient care should be transparently communicated, and moral distress should be acknowledged as treatment and care responsibility is shared across teams. Intentionally creating these reflective and supportive spaces that center around the experience of care providers (e.g., Balint groups) is key as clinicians strive to deliver the best care for their patients.

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In this moment of profound strain, the path forward lies not in denying the hardships facing clinicians and trainees but in meeting them with clarity, solidarity, and purpose. Below, we offer suggestions for navigating these tumultuous times, using our humanistic values as the central guiding principle, a “True North”.

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Provide historical perspective. As reflected in their responses to epidemics, new illnesses, wartime medicine, and scarcity of resources, most physicians, be they learners or teachers, ultimately navigate a path out of darkness. Current dilemmas are shaped by context and systemic challenges – not an individual failing on any one trainee’s part⁸⁹⁻¹¹⁰. Current trainees may not have this longitudinal perspective and would likely benefit from historical recollections and narratives for assurance and reminders that “this too shall pass.” In the absence of this historical anchoring, modeling resilience, and authentic disclosure on the part of mentors, trainees remain vulnerable to the fallout of their own high expectations. Some trainees even seek out geographical relocation to escape the angst of value misalignment.

~~Focus on the best interests of the patient. Our humanistic ideals remain critical, central, and immutable, as is our duty to uphold them even in the face of the current political and geopolitical climate. A foundation of universal values, which requires institutional/organizational support, can foster clinicians’ ability to lead with ethical, professional, and moral integrity even during challenging circumstances¹¹. Although it appears intuitive, trainees need explicit reminders to take care of the patient in front of you, do what you can, don’t dwell on what you can’t.~~

Acknowledge the uncertainty of our times, validate the stress it generates, and balance the emotional toll it brings with hope. The history of medicine is rich with examples of resilience. An honest appraisal of current barriers and the relentless nature of recent events that have

inhibited recovery of stressed healthcare systems is needed. This conception must be balanced with hope that is empirically grounded in the history of medicine: many such 'unprecedented times' have occurred, and clinicians have persevered through them. Creating a community that upholds the tenets of psychological safety can foster networks and connections in times when trainees feel alone and isolated¹². Faculty can create a safe space for learners to reflect on challenges, process their emotional responses to difficult experiences, explore perceived limitations, and debrief when current barriers to care adversely impact the patient. A safe space also promotes well-being and the sense of being valued.

Senior clinicians should model and articulate how they are preserving their values and ideals in current times. Mentors and educators are essential for guiding and shaping the careers of future clinicians, understanding their growth from a developmental and trauma-aware perspective and fostering professional identity formation, especially as it is impacted by the current learning environment^{13,14}. ~~Senior clinicians should model and articulate how they are preserving their values and ideals in current times.~~ Authentically sharing historical perspectives, past experiences, and current challenges in decision making and learning can reduce the sense of isolation that trainees may feel that they are the only ones struggling with these conflicted feelings. Faculty should demonstrate professionalism, and that ethical practice and principles of patient-centered care remain central to day-to-day clinical decision making. Such modeling can be uplifting and even inspiring¹⁵.

~~Attend to Organizational Cultures. While individual resilience and mentorship are indispensable, they cannot flourish in a vacuum. The emotional labor expected of clinicians and~~

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trainees today is inseparable from the environment in which they work. Research suggests that change at the organizational level is, at a minimum, equally as important as fostering clinicians' individual humanistic attributes¹⁵. Humanistic values may guide personal practice, but without institutional structures that reinforce, rather than undermine, those values, even the most dedicated clinicians will struggle to sustain them. Thus, any effort to safeguard trainee development and patient-centered care must extend beyond the dyad of learner and mentor to encompass the broader systems that shape clinical training.

Discuss the historical evolution of organizational cultures in academic medicine.

Throughout history, the most transformative growth has resulted when structural changes at organizational level have fostered the clinician's ability to deliver humanistic care under extreme duress. Lessons learned from wartime and pandemic crises that led to forced systemic solutions highlight the importance of supportive and communicative organizational cultures rather than individual fortitude.¹⁶ While individual resilience and mentorship are indispensable, they cannot flourish in a vacuum. The emotional labor expected of clinicians and trainees today is inseparable from the environment in which they work. Cultivating these humanistic attributes in organizations is equally important as in individuals to counter the negative effects of healthcare evolution.¹⁷

Emphasize the critical importance of institutional support to maintain the patient-provider alliance. Institutional support is critical to ensure faculty and trainees are protected in doing right by their patients.

Support from leadership can range from regular check-ins individually to debriefing as a group with a focus on promoting moral agency, teamwork, and resilience-building activities. Despite external pressures and the outward-facing measures they

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must take, it is incumbent on institutions to ensure that institutional values remain aligned with and committed to patient-centered care.

To translate these conceptual values into practice, **Table 1** offers some practical guidance and strategies that can be translated and tailored across specialties.

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In Conclusion

Medicine has always evolved through adversity. Even in these uncertain times, with the guiding principle of ~~north star~~ of patient-centered care and a community united around this principle, we can both withstand and transcend the pressures of the present.

References:

1. Singer T, Klimecki OM. Empathy and compassion. *Curr Biol*. 2014;24(18):R875-R878.
2. Graves J, Joyce C, Ahmed M, Hegazi I. Compassion fatigue in medical students and recent medical graduates: a scoping review. *Teach Learn Med*. 2025;;1-20.
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6. Mangurian C, Fitelson E, Devlin M, Pumar M, Epel E, Dahiya P, et al. Envisioning the future of well-being efforts for health care workers—successes and lessons learned from the COVID-19 pandemic. *JAMA Psychiatry*. 2023;80(9):962-967.

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7. Hughes MT, Rushton CH. Ethics and well-being: the health professions and the COVID-19 pandemic. *Acad Med*. 2022;97(3S):S98-S103.

8. Rider EA, Kurtz S, Slade D, Longmaid HE III, Ho MJ, Pun JK, et al. The International Charter for Human Values in Healthcare: an interprofessional global collaboration to enhance values and communication in healthcare. *Patient Educ Couns*. 2014;96(3):273-280.

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9. McLean M, Jha V, Sandars J. Professionalism under fire: conflict, war and epidemics. *Med Teach*. 2015;37(9):831-836.

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10. Lerner BH, Caplan AL. Judging the past: how history should inform bioethics. *Ann Intern Med*. 2016;165(9):675.

11. Coria AL, Rabin TL, Rule AR, Haq H, Hudspeth JC, Ratner L, et al. Global health crisis, global health response: how global health experiences prepared North American physicians for the COVID-19 pandemic. *J Gen Intern Med*. 2022;37(1):217-221.

11. Rider EA, Kurtz S, Slade D, Longmaid HE III, Ho MJ, Pun JK, et al. The International Charter for Human Values in Healthcare: an interprofessional global collaboration to enhance values and communication in healthcare. *Patient Educ Couns*. 2014;96(3):273-280.

12. Edmondson AC. *The Fearless Organization: Creating Psychological Safety in the Workplace for Learning, Innovation, and Growth*. Hoboken, NJ: John Wiley & Sons; 2018.
13. Whitaker RC, Payne GB, O'Neill MA, Brennan MM, Herman AN, Dearth-Wesley T, et al. Trauma-informed undergraduate medical education: a pathway to flourishing with adversity by enhancing psychological safety. *Perspect Med Educ*. 2024;13(1):324.
14. Toubassi D, Schenker C, Roberts M, Forte M. Professional identity formation: linking meaning to well-being. *Adv Health Sci Educ Theory Pract*. 2023;28(1):305-318.
15. Jones DS, Greene JA, Duffin J, Warner JH. Making the case for history in medical education. *J Hist Med Allied Sci*. 2015;70(4):623-652. doi:10.1093/jhmas/jru026
16. Curtin M, Richards HL, Fortune DG. Resilience among health care workers while working during a pandemic: a systematic review and meta-synthesis of qualitative studies. *Clin Psychol Rev*. 2022.
17. Rider EA, Gilligan MC, Osterberg LG, Litzelman DK, Plews-Ogan M, Weil AB, et al. Healthcare at the crossroads: the need to shape an organizational culture of humanistic teaching and practice. *J Gen Intern Med*. 2018;33(7):1092-1099.
18. Bryan CS, Longo LD. Perspective: teaching and mentoring the history of medicine: an Oslerian perspective. *Acad Med*. 2013.
19. Teo R, Grosser R, Thuppal H, Statter MB. "Sip & Share": building resilience in surgery residency through moral distress rounds. *J Surg Educ*. 2023.
20. Metselaar S, Molewijk B. Fostering moral resilience through moral case deliberation. *Nurs Ethics*. 2023.

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Formatted: English (United States)

21. Feingold JH, Kaplan CA, Hart A, et al. We Get by With a Little Help From Our PEERS: The Practice Enhancement, Engagement, Resilience, and Support Program for Building Community and Well-Being in Medical Education. *Acad Med*. 2022;97(6):858-862. doi:10.1097/ACM.0000000000004669

Formatted: English (United States)

22. Horvath AJ, Shegufta FA, Chairilsyah MR, Wikanta K, Waidyatillake NT, Reid K. The effectiveness of interventions designed to enhance and maintain medical student wellbeing: a systematic review. *BMC Med Educ*. 2026;26(1):667. Published 2026 Mar 7. doi:10.1186/s12909-025-08460-5

Formatted: English (United States)

23. Thomas LR, Brigham T, Shanafelt T. Residency and Fellowship: Fostering Physician Well-being Over the Career Life Cycle. *Mayo Clin Proc*. 2025;100(9):1649-1659. doi:10.1016/j.mayocp.2025.05.024

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Training Through Turbulence: Why Medical Education Must Build Intentionally Supportive, Historically Grounded Learning Cultures

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Dear Editors,

Please find attached our Perspective manuscript, “Training Through Turbulence: Why Medical Education Must Build Intentionally Supportive Learning Cultures,” submitted for consideration in the Journal of Graduate Medical Education.

Medical education is occurring amid prolonged turbulence marked by post-pandemic strain, moral distress influenced by current political policies, and systemic constraints that challenge trainees’ empathy, well-being, and professional identity formation. In this manuscript, we argue that addressing these challenges requires intentionally supportive learning cultures—beyond individual resilience—grounded in humanistic values, mentorship, psychological safety, and institutional alignment. Drawing on historical perspective and educator experience, we outline how medical educators and organizations can help trainees sustain integrity, purpose, and patient-centered care in uncertain times. This is the challenge of the current moment, and timely given what we are hearing from our trainees.

The manuscript adheres to the journal’s word limits and formatting requirements. This work has not been previously published in print or electronic form and is not under consideration by any other journal or publication.

Thank you for your consideration. We appreciate the opportunity to submit our work to JGME and look forward to your response.

Sincerely,

Zheala Qayyum, MD, MMSc